

HANDS-ON LAB EXERCISE 02

Git Branching and Pull Request Workflow

Training Information

Training Name: Git Branching and Pull Request Workflow

Trainer: Sandeep Kumar Sharma

Duration: 90 Minutes

Difficulty Level: Beginner to Intermediate

Lab Type: Hands-On Practical Exercise

Learning Objective

By the end of this lab, participants will be able to:

- Understand Git Branching concepts
 - Create and switch branches
 - Push branches to GitHub
 - Create Pull Requests (PR)
 - Review code changes
 - Merge Pull Requests
 - Delete branches after merge
 - Understand real-world developer collaboration workflow
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Learning Outcome

After completing this lab, participants will be able to:

- Work with multiple Git branches
 - Develop features independently
 - Raise Pull Requests
 - Review code before merging
 - Merge code safely into the main branch
 - Follow industry-standard GitHub workflow
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Prerequisites

Participants must complete:

- GitHub Fundamentals Lab
 - Git Installation
 - GitHub Repository Creation
 - Git Push and Pull Operations
-

Introduction

What is a Branch?

A branch is an independent line of development.

Developers create branches to work on new features without affecting the main code.

Example:

```
main
|
+---- feature-login
|
+---- feature-payment
|
+---- feature-search
```

Why Do We Need Branches?

Without branches:

- Multiple developers may overwrite each other's code.
- Production code can become unstable.

With branches:

- Developers work independently.
 - Code is reviewed before merging.
 - Production remains stable.
-

Real World Pull Request Workflow

```
Main Branch
|
Create Feature Branch
|
Develop Feature
|
Commit Changes
|
Push Branch
|
Create Pull Request
|
Code Review
|
Approve Changes
|
Merge Pull Request
|
Delete Feature Branch
```

Exercise 1: Clone Existing Repository

Open Terminal.

```
git clone https://github.com/username/github-lab.git
```

Move into repository.

```
cd github-lab
```

Check current branch.

```
git branch
```

Expected Output:

```
* main
```

Exercise 2: Create Feature Branch

Create new branch.

```
git branch feature-login
```

Verify:

```
git branch
```

Expected Output:

```
* main  
feature-login
```

Exercise 3: Switch to Feature Branch

```
git checkout feature-login
```

OR

```
git switch feature-login
```

Verify:

```
git branch
```

Expected Output:

```
* feature-login  
main
```

Exercise 4: Create New Feature

Open:

```
index.html
```

Add content:

```
<h2>Login Feature Added</h2>
```

Save file.

Exercise 5: Commit Feature Changes

Check status.

```
git status
```

Add files.

```
git add .
```

Commit changes.

```
git commit -m "Added Login Feature"
```

Exercise 6: Push Feature Branch

Push branch to GitHub.

```
git push -u origin feature-login
```

Expected Output:

```
Branch 'feature-login' set up to track remote branch.
```

Refresh GitHub repository.

You will notice a new branch.

Exercise 7: Create Pull Request

Login to GitHub.

Open repository.

GitHub will display:

Compare & Pull Request

Click:

Compare & Pull Request

Exercise 8: Configure Pull Request

Provide:

Title

Added Login Feature

Description

This pull request adds login functionality to the application.

Click:

Create Pull Request

Exercise 9: Review Pull Request

Observe:

- Files Changed
- Commits
- Conversation Tab

Review code modifications.

Verify:

```
index.html changes visible
```

Exercise 10: Approve and Merge Pull Request

Click:

```
Merge Pull Request
```

Then:

```
Confirm Merge
```

Expected Result:

```
Pull Request Successfully Merged
```

Exercise 11: Verify Merge

Switch to Main Branch.

```
git checkout main
```

Pull latest code.

```
git pull origin main
```

Open:

```
index.html
```

Verify:

```
<h2>Login Feature Added</h2>
```

is available.

Exercise 12: Delete Feature Branch

Delete local branch.

```
git branch -d feature-login
```

Delete remote branch.

```
git push origin --delete feature-login
```

Verify:

```
git branch
```

Expected Output:

```
* main
```

Important Commands

Command	Description
git branch	View Branches
git branch branch-name	Create Branch
git checkout branch-name	Switch Branch
git switch branch-name	Switch Branch
git add .	Stage Changes
git commit -m	Commit Changes
git push -u origin branch-name	Push Branch
git pull	Get Latest Changes
git branch -d	Delete Local Branch
git push origin --delete	Delete Remote Branch

Challenge Activity

Perform the following independently:

1. Create branch feature-payment
 2. Add Payment Page
 3. Commit changes
 4. Push branch
 5. Create Pull Request
 6. Merge Pull Request
 7. Pull latest code
 8. Delete branch
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Lab Validation Checklist

Task	Completed
Repository Cloned	☐
Feature Branch Created	☐
Branch Switched	☐
Code Modified	☐
Commit Created	☐
Branch Pushed	☐
Pull Request Created	☐
Pull Request Reviewed	☐
Pull Request Merged	☐
Branch Deleted	☐

Lab Summary

In this lab, participants learned:

- Git Branching
- Feature Development Workflow
- Pull Request Creation
- Code Review Process
- Pull Request Merge Process
- Branch Cleanup Activities

This workflow is used by almost every software development team and forms the foundation of collaborative development using GitHub.

Trainer: Sandeep Kumar Sharma

Lab Status: Successfully Completed 